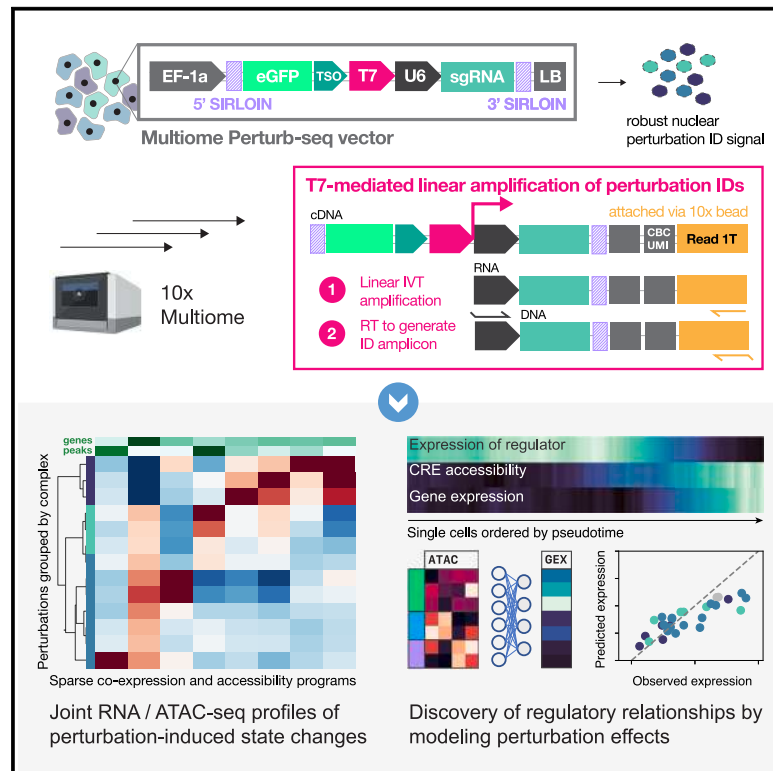


Cell Systems

Multiome Perturb-seq unlocks scalable discovery of integrated perturbation effects on the transcriptome and epigenome

Graphical abstract



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In brief

Multiome Perturb-seq extends single-cell CRISPR screens to simultaneously quantify perturbations' effects on the epigenome and transcriptome. A CRISPRi screen of chromatin remodelers using this method revealed distinct effects on each aspect and enabled the modeling of the regulatory relationship between chromatin accessibility and gene expression.

Highlights

- Single-cell CRISPR screen simultaneously measuring gene expression and chromatin state
- Engineered sgRNA expression vector and library prep compatible with commercial reagents
- Modeling perturbations' effects connects regulatory elements to the genes they control



Methods in Brief

Multiome Perturb-seq unlocks scalable discovery of integrated perturbation effects on the transcriptome and epigenome

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SUMMARY

Single-cell CRISPR screens link genetic perturbations to transcriptional states, but high-throughput methods connecting these induced changes to their regulatory foundations are limited. Here, we introduce Multiome Perturb-seq, extending single-cell CRISPR screens to simultaneously measure perturbation-induced changes in gene expression and chromatin accessibility. We apply Multiome Perturb-seq in a CRISPRi screen of 13 chromatin remodelers in human RPE-1 cells, achieving efficient assignment of sgRNA identities to single nuclei via an improved method for capturing barcode transcripts from nuclear RNA. We organize expression and accessibility measurements into coherent programs describing the integrated effects of perturbations on cell state, finding that *ARID1A* and *SUZ12* knockdowns induce programs enriched for developmental features. Modeling of perturbation-induced heterogeneity connects accessibility changes to changes in gene expression, highlighting the value of multimodal profiling. Overall, our method provides a scalable and simply implemented system to dissect the regulatory logic underpinning cell state. A record of this paper's transparent peer review process is included in the supplemental information.

INTRODUCTION

Single-cell CRISPR screens have transformed the field of functional genomics by enabling simple, pooled experiments linking genetic perturbations to high-content phenotypes.^{1,2} Perturb-seq, CROP-seq, and Mosaic-seq pioneered these approaches, expanding screen readouts from scalar measurements of cell fitness or reporter expression to comprehensive assessment of changes in transcriptional state.^{3–6} As single-cell assays have advanced, techniques profiling multiple modalities have integrated transcriptional data with relevant covariates,^{7–10} enabling new questions to be asked about the origins and regulation of cellular states. In particular, multiome assays that simultaneously measure gene expression and chromatin accessibility in the same cells have the potential to help explain how chromatin state influences and directs the transitions between transcriptional states.^{11,12} During development, changes in accessibility have been shown to “foreshadow” gene expression changes and ultimately specify cell lineages.^{11,13–15} Conversely, bulk co-profiling of expression and accessibility changes following chemical perturbations has suggested that certain pathways induce expression changes within open chromatin independent of remodeling, whereas others trigger global rewiring of chromatin state.^{12,16–18} The unclear interplay between these distinct steps in gene expres-

sion calls for a method to simultaneously characterize many perturbation-induced effects on the transcriptome and epigenome in parallel. These connections would have diverse applications because of their mechanistic interpretability, from elucidating regulators of cellular plasticity to gathering data to constrain causal modeling of gene regulatory networks.

To this end, we introduce Multiome Perturb-seq, an extension of the Perturb-seq platform that captures simultaneous single-cell RNA sequencing (scRNA-seq) and scATAC-seq measurements of perturbation effects. Our protocol adapts the CROP-seq vector⁵ to be compatible with commercially available 10x Single Cell Multiome ATAC + Gene Expression reagents (“10x Multiome”). The core challenge is robust sgRNA assignment from nuclear cDNA, as isolating nuclei is a requirement of scATAC-seq protocols. To overcome this, we modified the CROP-seq vector and developed a new amplification protocol to enhance recovery of CROP-seq sgRNA identity transcripts (“IDs”), increasing perturbation assignment rate to 68%. We apply this protocol to a library of 13 chromatin remodelers from three regulatory complexes, demonstrate how Multiome Perturb-seq can profile and organize expression and accessibility changes into coherent effects on cell state, and interpret the dynamic regulatory processes connecting these two layers of gene regulation.



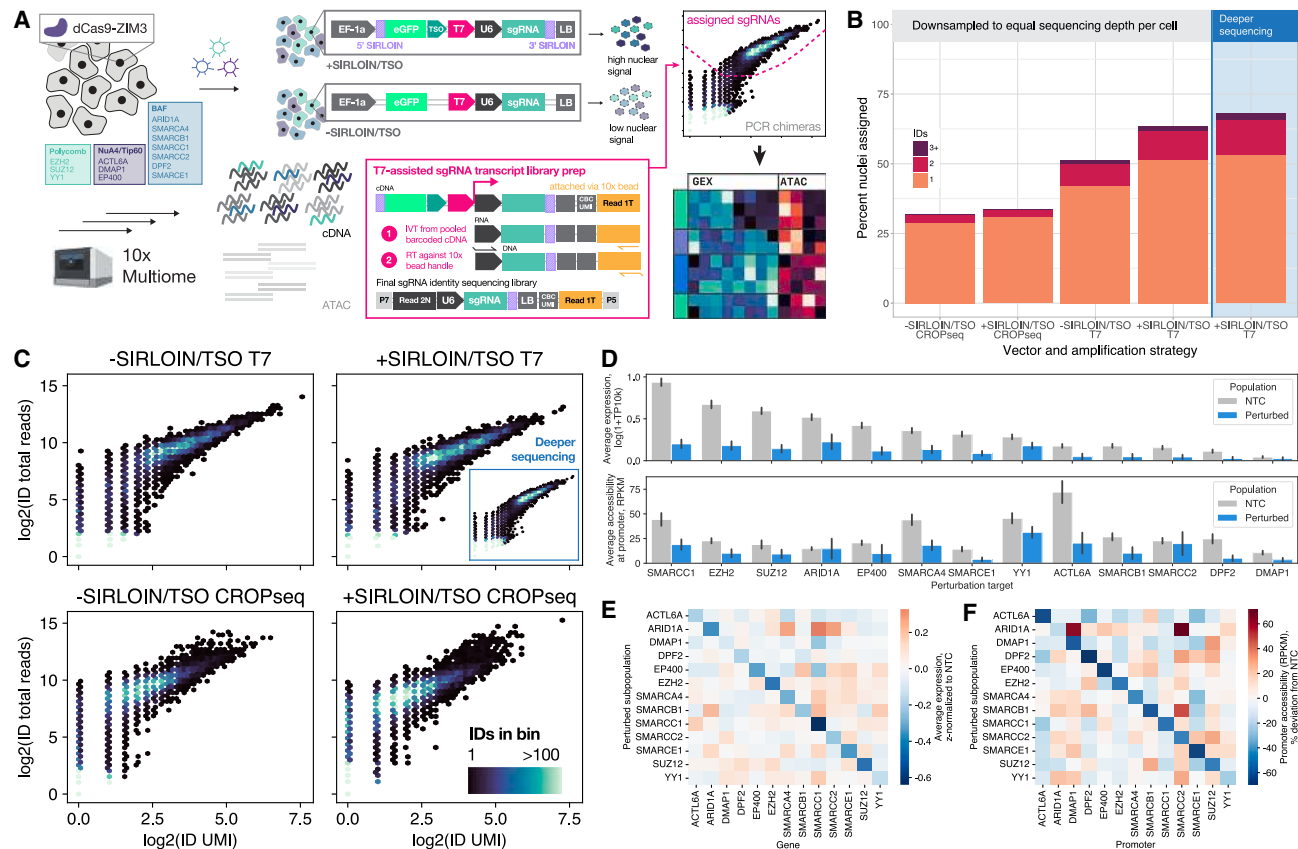


Figure 1. Improved perturbation assignment to single nuclei with T7-assisted sgRNA library prep and an engineered CROP-seq vector

(A) Schematic of Multiome Perturb-seq experimental design, library, and guide assignment strategy. sgRNA identity transcripts are biased toward nuclear retention by addition of dual SIRLOIN elements flanking the sgRNA cassette (purple), and a partial 10x TSO capture sequence (green) assists recovery. Transcripts are linearly amplified via an internal high-activity T7 promoter from barcoded cDNA. U6, modified U6 Pol III promoter, LB: lineage barcode cassette.

(B) Percentage of single nuclei assigned to 1, 2, or 3+ sgRNA identities using traditional CROP-seq or T7-assisted library prep, in a standard CROP-seq vector with T7 promoter (-SIRLOIN/TSO) or a CROP-seq vector with T7 promoter, dual SIRLOINs and partial TSO capture sequence (+SIRLOIN/TSO), at indicated sequencing depths (downsampled for comparison, deeper sequencing for downstream analysis).

(C) Density of UMI and total sequencing reads of sgRNA identity transcripts amplified by traditional CROP-seq or T7-assisted library prep in nuclei-containing droplets, in each vector condition, downsampled to equal sequencing depth per nucleus.

(D) Average single-cell expression and promoter accessibility of perturbation target genes in NTC-assigned populations vs. populations assigned to each perturbation. Error bars show 95% confidence intervals.

(E) Average normalized expression of perturbation target genes in perturbed subpopulations.

(F) Pseudobulk normalized accessibility at promoters of perturbation target genes in perturbed subpopulations.

RESULTS

An optimized sgRNA assignment workflow recovers perturbation identities of single nuclei

We chose 10x Multiome as the target platform due to its ubiquity, quality, and the commercial availability of reagents, but this introduces a challenge for sgRNA capture: the corresponding beads contain oligo-dT primers to capture mRNA and ATAC primers to capture transposed genomic DNA but critically lack primers for capture of sgRNAs. This limitation restricted our ID expression and assignment strategy to the polyadenylated transcript-dependent CROP-seq approach. Accurate assignment is thus hindered by both the lower transcript count in isolated nuclei and the active nuclear export of polyadenylated mRNAs.¹⁹

To mitigate this effect, we engineered a modified CROP-seq vector and library preparation protocol to increase capture and

assignment of sgRNA ID transcripts (Figure 1A). We incorporated an internal high-activity T7 promoter²⁰ upstream of the sgRNA cassette (similarly to methods used to boost sgRNA signal in spatial contexts²¹), enabling efficient linear amplification of IDs from cDNA before index PCR (STAR Methods, Methods S1). We also flanked the ID transcript with short sequence elements (“SIRLOINs”) that increase nuclear localization of RNA,²² added a partial 10x TSO (template switching oligonucleotide) binding sequence to assist amplification of IDs in case of incomplete template switching during reverse transcription,²³ and removed the highly structured woodchuck hepatitis virus post-transcriptional regulatory element (WPRE), which we reasoned might impair efficiency of reverse transcription. We observed that SIRLOIN elements drove a modest increase in nuclear abundance of IDs coupled with a modest decrease in overall expression, consistent with their original characterization

(Figure S1A), and did not alter knockdown efficiency of cell surface markers (Figure S1B). Finally, we included a cassette for inserting auxiliary barcodes (e.g., as lineage markers^{24,25}), which can be read out alongside IDs via altered sequencing parameters in our protocol if desired (Methods S1).

We applied Multiome Perturb-seq in a CRISPRi screen of retinal pigment epithelial (RPE-1) cells stably expressing ZIM3-dCas9,²⁶ using both the fully engineered Multiome Perturb-seq vector and a vector lacking SIRLOIN and TSO elements. Importantly, these hTERT-immortalized but untransformed cells allow us to investigate gene regulation in a more physiological context than cancer cell lines. Our library included three non-targeting control (NTC) sgRNAs and 13 sgRNAs targeting chromatin remodelers belonging to three regulatory complexes^{27–30} (Figure 1A; Table S1). We isolated nuclei from ZIM3-dCas9 RPE-1 populations expressing each version of the library, encapsulated them in two 10x Multiome lanes targeting 10,000 nuclei each, and recovered 9,318 (+SIRLOIN/TSO) and 9,504 (–SIRLOIN/TSO) nuclei after cellranger-arc cell calling (STAR Methods).

At equal sequencing depth of 850 reads per nucleus, traditional CROP-seq PCR-based library amplification yielded 31.8% assignment of single nuclei to at least one ID, improving to 33.8% in the +SIRLOIN/TSO condition (Figures 1B and S2). By contrast, our T7 amplification protocol yielded 51.2% assignment in the –SIRLOIN/TSO condition, increasing to 63.4% in the +SIRLOIN/TSO condition. Deeper sequencing of the T7 –SIRLOIN/TSO library to ~4,000 reads per nucleus modestly increased assignment to 68.0%. We traced these efficiency differences to shifts in library composition. Linear amplification of IDs before PCR shifted the final number of ID unique molecular identifiers (UMIs) and total read density toward the high-UMI/high-read mode representing assignable transcripts, with a more pronounced shift in the +SIRLOIN/TSO condition (Figure 1C, STAR Methods). Greater assignment efficiency also allowed more sensitive filtering of multiple assignments (multiple infections or cells co-encapsulated within droplets) that would otherwise confound analysis. Recovery of ID UMIs varied minimally among perturbation assignments (Figure S3A). Our assignment rate also matched the rate we observed for direct capture Perturb-seq³¹ performed on isolated nuclei using the 10x 3' Gene Expression chemistry, which captures sgRNAs directly instead of ID transcripts (Figure S3B). Overall, our ID assignment workflow represents a greater-than-40% improvement over the current state-of-the-art single-cell ATAC-seq screening approach on the 10x platform (spear-ATAC: 48% assignment, Chromium Single Cell ATAC v1),³² while additionally providing gene expression phenotypes.

Analysis of the +SIRLOIN/TSO lane, after filtering for single ID assignments and performing basic quality control (Figure S4; Table S2; STAR Methods), yielded a final population of 4,724 nuclei with an average of 2,739 genes and 19,202 unique ATAC fragments detected. We observed robust on-target knockdown of expression and decreased accessibility at target gene promoters in perturbed populations (Figure 1D). Notably, we observed these effects across a range of native chromatin contexts, with both high-accessibility (e.g., *ACTL6A*) and low-accessibility (e.g., *DMAP1*) promoters displaying decreased accessibility upon perturbation. This result is consistent with

the mechanism of the ZIM3 CRISPRi effector, whose KRAB domain recruits repressive chromatin-modifying complexes to induce heterochromatin formation.^{26,33} Although the on-target effect of each perturbation was strongest, as expected, we additionally observed modest upregulation of *SMARCA4*, *SMARCC1*, and *SMARCC2* in the sgARID1A-containing population, suggesting crosstalk between these BAF complex members (Figure 1E). Increased expression of *SMARCC2* was concordant with increased accessibility of the *SMARCC2* promoter in sgARID1A-containing cells (Figure 1F). Overall, our optimized assignment workflow efficiently associates perturbations with single-cell transcriptomes and accessibility profiles.

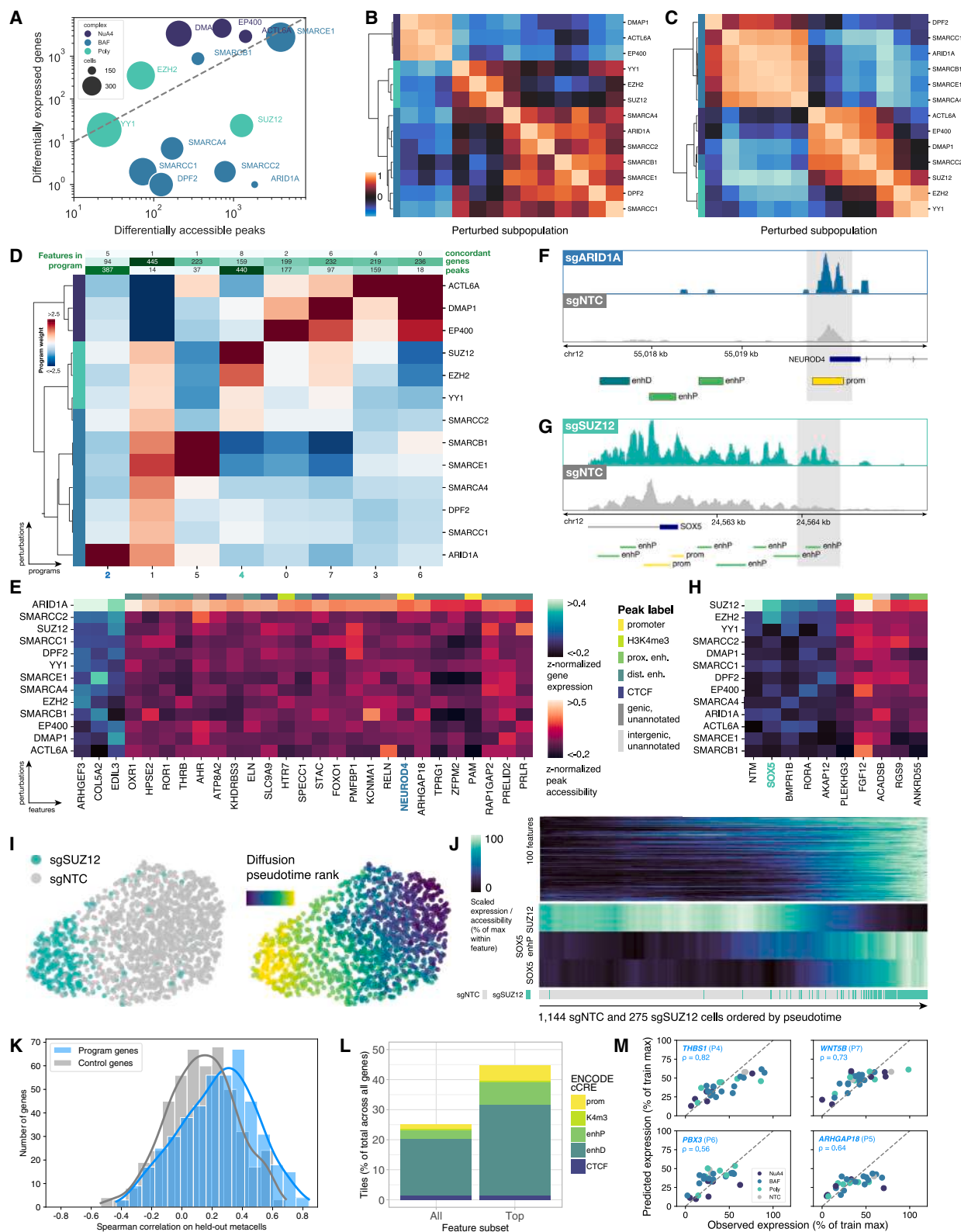
Induced accessibility and expression changes provide complementary and largely non-overlapping descriptions of perturbation effects

We next analyzed the global phenotypes induced by perturbations. Identification of differentially expressed genes and differentially accessible peaks within each perturbation (Table S3; STAR Methods) revealed highly variable behaviors among individual perturbations: some mainly affected the chromatin landscape (*ARID1A* and *SMARCC2*), whereas others mostly modified gene expression (*DMAP1* and *EP400*), and some altered both equally (*SMARCE1* and *YY1*) (Figure 2A). In line with previous work,¹⁶ we found that although examples of concordant changes in accessibility and expression at loci other than perturbation targets were present in 7 of 13 perturbations, these instances were relatively rare, and most differential expression and accessibility events across perturbations were independent (Table S4).

To compare how the distinct readouts organized perturbations, we performed hierarchical clustering on both expression and accessibility profiles (STAR Methods). Both modalities separated targets into known complexes, with tighter clustering mirroring the relative impact of different complexes on either the transcriptome or chromatin state: NuA4 members clustered distinctly in transcriptome space, whereas BAF complex members were clearly delineated in accessibility space (Figures 2B and 2C). Similarly, UMAP representations of each individual modality showed the modest clustering by perturbation typical of Perturb-seq experiments, but a joint UMAP embedding of expression and accessibility displayed greater clustering by complex membership (Figure S5). These differences underscore that expression and accessibility profiles offer complementary information, enabling deeper characterization of perturbation effects across related targets.

Co-embedding genes and peaks into sparse programs organizes effects on cell state

To organize the effects of perturbations into coherent programs, we concatenated filtered expression and accessibility matrices and co-embedded genes and peaks via non-negative sparse principal-component analysis (PCA) (STAR Methods). This approach, applied to 7,885 genes and 5,060 peaks differentially expressed or accessible in at least one perturbation (and for peaks, limited to those within 50 kb of a gene), yielded 8 sparse programs, each containing hundreds of individual features (Figure 2D; Table S5). Notably, programs varied in composition: some



(legend on next page)

programs (2 and 4) were largely peak-driven, others (1, 5, and 6) gene-driven, and others (0, 3, and 7) mixed. Labeling each peak with its nearest gene to facilitate interpretation, we focused on programs 2 and 4, reasoning that these represented more consequential rewiring of cell state than gene-driven programs operating within open chromatin.

Program 2, which was highly specific to the sgARID1A-containing population, featured multiple marker genes of photoreceptors and retinal progenitors from an *in vivo* atlas of retinal development³⁴ (Figure 2E). Mapping the program back onto the atlas also showed concentrated expression within these cell types (Figure S6; STAR Methods), suggesting it may reflect the reactivation of a gene expression program active during retinal development and the known epigenetic plasticity of RPE cells.^{35–37} Further supporting this hypothesis, we observed increased accessibility at the promoter of *NEUROD4*, a transcription factor (TF) normally expressed in retinal progenitors throughout embryonic development³⁸ that specifies photoreceptor differentiation and function³⁹ (Figure 2F). Although *ARID1A* knockdown did not directly induce *NEUROD4* expression, the increased promoter accessibility suggests that it may be primed for activation by additional stimuli, hinting at a synergistic role for *ARID1A* and *NEUROD4* in cell fate determination during retinal development.

Modeling perturbation-induced heterogeneity of expression and accessibility elucidates regulatory relationships

The single-cell resolution of Multiome Perturb-seq has the potential to yield deeper insights into the relationship between chromatin accessibility and gene expression. In program 4, specific to Polycomb members *EZH2* and *SUZ12*, we found high feature weights on *SOX5* (a TF implicated in nervous system and skeletal system development^{40,41}). We observed a global increase in accessibility around the *SOX5* TSS (transcription start

site) as well as a differentially accessible peak corresponding to a *SOX5* proximal enhancer in sgSUZ12-containing cells (Figure 2G), and program 4 contained several *SOX5* target genes (Figure 2H).

Given these concordant changes, we reasoned that sgSUZ12-containing cells could anchor a pseudotime trajectory ordering heterogeneity within both control and perturbed populations, allowing us to visualize sequential chromatin opening preceding increased gene expression.^{42–44} As expected, a diffusion pseudotime metric constructed on a combined population of 1,419 sgSUZ12-containing and sgNTC-containing cells (STAR Methods) mapped a trajectory from control cells to perturbed cells (Figure 2I). We visualized the expression and accessibility of the top 100 features by weight within program 4 in single cells ordered by pseudotime, revealing broad upregulation aligned with this principal axis. (Figure 2J, top) Some features “fired” earlier along the trajectory than others, suggesting candidate regulatory relationships between these “early” features and “late” targets. In the case of *SOX5*, accessibility of the proximal enhancer precedes increased *SOX5* expression and mirrors *SUZ12* knockdown, suggesting a causal role for *SUZ12* in repression of this locus and for this proximal enhancer in modulating the expression level of *SOX5* (Figure 2J, bottom).

Finally, we searched more globally for regulatory relationships revealed by our perturbations. Previous methods have demonstrated the power of multiome data to map *cis*-regulatory elements by modeling how differences in chromatin accessibility predict differences in gene expression across cell types.^{45,46} We reasoned that perturbation-induced changes would allow the same type of modeling within a cell type. We trained ensemble models with AutoGluon⁴⁷ to predict gene expression as a function of chromatin accessibility within 250 kb of the gene body. We applied this approach to 457 protein-coding genes with high feature weights in our programs, as well as to 457 non-differential, expression-matched “control” genes (STAR Methods). Predictive

Figure 2. Multiome Perturb-seq characterizes transcriptional and epigenetic state to elucidate regulatory relationships

- (A) Number of differentially expressed genes ($q < 0.1$, Kolmogorov-Smirnov test on normalized expression, Benjamini-Hochberg false discovery rate [BH FDR] correction) vs. number of differentially accessible peaks ($q < 0.1$, Mann-Whitney test on paired insertion counts within peaks, BH FDR correction) for each perturbation target. Point sizes represent number of cells assigned each sgRNA identity.
- (B) Hierarchical clustering of average PCA embeddings of normalized expression in perturbed subpopulations.
- (C) Hierarchical clustering of average spectral embeddings of ATAC tile matrix in perturbed subpopulations; 2.5×10^5 tile features selected based on accessibility across all cells.
- (D) Program weights on perturbed subpopulations and number of features with nonzero loadings per program from sparse PCA embedding of 7,885 genes and 5,060 peaks differentially expressed or accessible in at least one perturbation condition.
- (E) Normalized expression and accessibility of marker genes of primary and neurogenic retinal progenitors and photoreceptors (from Zuo et al.³⁴) in program 2; peaks labeled with nearest gene.
- (F) Pseudobulk ATAC-seq track at the *NEUROD4* promoter in sgARID1A-containing and sgNTC-containing cells, with ENCODE cCRE annotations (prom, promoter; enhP, proximal enhancer; enhD, distal enhancer). Differential peak selected as feature in program 2 highlighted in gray.
- (G) Pseudobulk ATAC-seq track at the *SOX5* promoter for sgSUZ12-containing and sgNTC-containing populations, with ENCODE cCRE annotations (prom, promoter; enhP, proximal enhancer; enhD, distal enhancer). Differential peak in proximal enhancer highlighted in gray.
- (H) Normalized expression and accessibility of *SOX5* and *SOX5* target genes in program 4.
- (I) UMAP representations of 1,419 sgSUZ12-containing and sgNTC-containing single cells, colored by sgRNA identity and diffusion pseudotime rank.
- (J) (Top) Expression and accessibility of top-100 program 4 features by weight in the population shown in Figure 2I. (Bottom) Expression of *SUZ12*, accessibility of the *SOX5* proximal enhancer shown in (G), and expression of *SOX5* in the same population. Single cells are ordered along the x axis by their diffusion pseudotime values, with each feature separately scaled to a range of 0–1.
- (K) Accuracy (test set Spearman correlation between observed and predicted normalized expression) of models predicting gene expression as a function of local accessibility. Plot compares results for 457 protein-coding genes with top-100 feature weights within programs to 457 expression-matched control genes not differentially expressed in any perturbation condition.
- (L) Enrichment of ENCODE candidate *cis*-regulatory elements among features selected by models of program genes. Plot compares top tiles as assessed by permutation shuffling to the background distribution of all chromatin tiles used for modeling.
- (M) Predicted vs. observed normalized expression for program genes involved in retinal development and function with highly predictive models.

performance was variable, indicating that accessibility is generally necessary but not sufficient to determine expression and supporting the value of joint profiling. However, models of genes altered by our perturbations were more accurate than those made for control genes (mean test set Spearman correlation 0.240 vs. 0.135, $p < 10^{-10}$, t test; [STAR Methods](#); [Figure 2K](#)). Perturbations thus add information absent in cells at rest. Modeling performance was robust to absolute expression level, demonstrating that this approach applies in a variety of regulatory contexts ([Figure S7](#)). Via feature attribution in the trained models, we could also map potentially causal *cis*-regulatory elements ([STAR Methods](#); [Figure S8](#)). Across all models, the most predictive chromatin tiles were strongly enriched for ENCODE cCRE annotations, suggesting the model is able to recover *cis*-regulatory relationships ([Figure 2L](#)). Genes implicated in retinal function⁴⁸ and markers of retinal neurons *in vivo*³⁴ were also modeled with high accuracy ([Figure 2M](#)), demonstrating the potential of this approach and of Multiome Perturb-seq to help understand physiologically meaningful phenotypes.

DISCUSSION

We developed Multiome Perturb-seq, a single-cell CRISPR screening method that efficiently assigns perturbation identities to simultaneous transcriptional and epigenomic phenotypes. Though we target use with robust, commercially available reagents, the protocol is likely simple to adapt to other workflows like combinatorial indexing.^{12,49–51} Our T7-assisted amplification protocol and engineered Multiome Perturb-seq vector recovered perturbation identities for 68% of input nuclei, a >40% improvement over comparable techniques for scATAC-seq.³² These advances may also support *in vivo* applications of conventional Perturb-seq that require nuclei isolation.⁵² Our method revealed that different perturbations distinctly alter chromatin accessibility and gene expression. One important consequence is that larger-scale Multiome Perturb-seq experiments may help characterize the many genes that are successfully perturbed without apparent transcriptional effects in existing large-scale Perturb-seq experiments.⁵³

However, the deeper strength of the approach lies in its ability to link perturbation-induced expression changes to their regulatory underpinnings—a unique capability of the multiome readout. Modeling gene expression as a function of local accessibility features nominates *cis*-regulatory elements that may be responsible for mediating those changes, without requiring direct perturbations of those elements as in other approaches.⁵⁴ Instead, we exploit synthetic heterogeneity created by perturbations to reveal regulatory relationships hidden in cells at rest. Leveraging this at the single-cell level, we could visualize sequential changes to chromatin state that foreshadowed changes in gene expression. Future screens employing our vector's lineage barcodes and measuring changes over time could be used to study longer timescales, including epigenetic heterogeneity and memory within clonal lineages.⁵⁵

Looking forward, we envision that Multiome Perturb-seq will not only enable experiments that advance basic biological discovery but also facilitate rational engineering of cell state. By simultaneously profiling the transcriptome and epigenome, this method provides a view of both a cell's current state and

potential future avenues for perturbation, as the epigenomic data can reveal genes newly primed for activation. Our observation of increased *NEUROD4* promoter accessibility in response to *ARID1A* knockdown exemplifies this concept, suggesting *NEUROD4* may be primed for activation and represent the next step in a defined regulatory path. Such insights could guide the design of synergistic perturbations, where an initial change in chromatin state sensitizes specific genes to subsequent interventions. This approach could direct programmed combinatorial genetic perturbations toward previously unreachable cell states, with relevance for both basic research and translational applications.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Thomas M. Norman (normanm@mskcc.org).

Materials availability

Plasmids generated in this study have been deposited at Addgene: pEM040 [Multiome Perturb-seq sgRNA expression vector] #224898, pEM047 [T7-assisted CROPseq sgRNA expression vector] #224899.

Data and code availability

Sequencing data have been deposited at SRA and are publicly available as of the date of publication. Accession numbers are listed in the [key resources table](#). All original code has been deposited at Zenodo and is publicly available as of the date of publication. DOIs are listed in the [key resources table](#). Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

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AUTHOR CONTRIBUTIONS

Conceptualization, E.M., K.M.S., and T.M.N.; methodology, E.M., K.M.S., and T.M.N.; formal analysis, E.M.; investigation, E.M. and K.M.S.; writing—original draft, E.M.; writing—review and editing, E.M., K.M.S., and T.M.N.; visualization, E.M.; supervision, T.M.N.; funding acquisition, T.M.N.

DECLARATION OF INTERESTS

T.M.N. is an author on US Patent No. 11,214,797B2, related to Perturb-seq.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
CD81 APC-H7, Clone JS-81	BD Biosciences	Cat#656154; RRID:AB_2870403
Bacterial and virus strains		
NEB Stable Competent E. coli (High Efficiency)	New England Biolabs	Cat#C3040H
Chemicals, peptides, and recombinant proteins		
HiFi DNA Assembly Master Mix	New England Biolabs	Cat#E2621
Hi-T4 DNA Ligase	New England Biolabs	Cat#M2622
HiScribe T7 High Yield RNA Synthesis Kit	New England Biolabs	Cat#E2050
Maxima H Minus First Strand cDNA Synthesis Kit	ThermoFisher Scientific	Cat#K1652
KAPA HiFi HotStart ReadyMix	Roche	Cat#07958935001
ProNex Size-Selective Purification System	Promega	Cat#NG2001
TransIT-LTI Transfection Reagent	Mirus	Cat#MIR2300
RNAclean XP Reagent	Beckman Coulter	Cat#A63987
Protector RNase inhibitor	Sigma	Cat#3335402001
BstXI	New England Biolabs	Cat#R0113S
Bipl	New England Biolabs	Cat#R0585S
eBioscience™ 7-AAD Viability Staining Solution	ThermoFisher Scientific	Cat#00-6993-50
Critical commercial assays		
10x Chromium Next GEM Single Cell Multiome ATAC + Gene Expression Reagent Kit	10x Genomics	PN-1000285; Document #CG000338
10x Chromium Next GEM Single Cell 3' Reagent Kit v3.1	10x Genomics	PN-1000269
Monarch PCR & DNA Cleanup Kit	New England Biolabs	Cat#T1030
Qiagen Spin Miniprep Kit	Qiagen	Cat#27104
Qiagen Plasmid Plus Midiprep Kit	Qiagen	Cat#12941
Deposited data		
Raw sequencing data from Multiome Perturb-seq screen	This paper	SRA: PRJNA1128171
Processed scRNA-seq data from retina atlas	Zuo et al. ³⁴	https://doi.org/10.1038/s41467-024-50853-5
Experimental models: Cell lines		
Human: hTERT RPE-1 dCas9-ZIM3	Replogle et al. ⁵³	https://doi.org/10.1016/j.cell.2022.05.013
Human: HEK293T LentiX	Takara	Cat#632180
Oligonucleotides		
See Table S6	IDT	N/A
Recombinant DNA		
Multiome Perturb-seq vectors, pEM040 and pEM047	This paper	Addgene #224898, 224899
CRISPRi chromatin remodelers libraries	This paper	N/A
psPAX2 plasmid	Addgene	Addgene #12260
pMD2.G plasmid	Addgene	Addgene #11259
Software and algorithms		
Code for analysis in this work	This paper	https://doi.org/10.5281/zenodo.14217682

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Cellranger-arc version 2.0.2	10x Genomics	https://www.10xgenomics.com/support/software/cell-ranger/latest
Cellranger version 7.1.0	10x Genomics	https://www.10xgenomics.com/support/software/cell-ranger/latest
perturbseq	Norman et al. ⁵⁶	https://github.com/thomasmaxwellnorman/perturbseq_demo/tree/master/perturbseq
Scanpy version 1.9.3	Wolf et al. ⁵⁷	https://doi.org/10.1186/s13059-017-1382-0 , https://scanpy.readthedocs.io/en/stable/index.html
SnapATAC2 version 2.6.0	Zhang et al. ⁵⁸	https://doi.org/10.1038/s41592-023-02139-9 , https://kzhang.org/SnapATAC2/
AutoGluon version 1.1.1	Erickson et al. ⁴⁷	https://doi.org/10.48550/arXiv.2003.06505 , https://auto.gluon.ai/stable/index.html
Other		
DMEM-F12	Gibco	Cat#11320033
Fetal bovine serum	VWR	Cat#97068-085
Penicillin-Streptomycin	Gibco	Cat#15140122
Hygromycin B	ThermoFisher Scientific	Cat#10687-010
Bovine serum albumin	Sigma	Cat#A9418

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Cell lines and lentiviral packaging

RPE-1-ZIM3 (derived from hTERT RPE-1 [ATCC CRL-4000; RRID:CVCL_4388]) cells were obtained from Replogle et al.⁵³ and maintained in DMEM:F12 media (Gibco, 11320033), 10% Fetal Bovine Serum (FBS) (VWR, 97068-085), 100U penicillin and 100 mg/mL streptomycin (Gibco, 15140122), and 0.01 mg/mL Hygromycin B (Thermo Fisher, 10687-010). LentiX 293T cells were purchased from Takara Bio (632180) and maintained in DMEM (Gibco #11965092), 10% Fetal Bovine Serum (FBS) (VWR, 97068-085), 100U penicillin and 100 mg/mL streptomycin (Gibco, 15140122). For lentiviral packaging, LentiX cells were grown to 70–80% confluence and co-transfected with library vector, transfer, and packaging plasmids psPAX2 and pMD2G using TransIT-LTI Transfection Reagent (Mirus, MIR 2300). 24 hr post-transfection, media was changed to DMEM + 10% BSA (Sigma #A9418). Virus was harvested by filtering collected media through a 0.45µm filter 48 hr post-transfection.

METHOD DETAILS

Construction of engineered CROP-seq vectors

Vector pEM040 (+SIRLOIN/TSO, Addgene #224898) was constructed sequentially from vector pKS010 (a CROP-seq style vector with GFP marker and lineage barcoding capabilities, modified from pBA950 [Addgene plasmid #122239], available upon request from [lead contact](#)). A high-activity T7 promoter with appropriate overlaps was ordered from IDT as a duplexed oligo (Table S6) and cloned into PpuMI-linearized pKS010 via HiFi assembly (NEB #E2621). Ultramer oligos containing the SIRLOIN consensus sequence²² and appropriate HiFi assembly overlaps were ordered from IDT (Table S6) and sequentially cloned into the 5' and 3' UTRs of the CROP-seq transcript, first via linearization with BsiWI (5' UTR) to create an intermediate vector, then via double digestion with Sall and BamHI (3' UTR), removing an internal Nextera Read 2 from the lineage barcode cassette in the parent vector. The partial TSO sequence was ordered from IDT as an Ultramer oligo and similarly cloned by HiFi assembly into an intermediate vector linearized with MluI. Vector pEM047 (−SIRLOIN/TSO, Addgene #224899) was constructed by adding the T7 promoter as described above and digesting with Sall / BamHI to remove the Nextera Read 2, but the 3' SIRLOIN oligo was replaced with a duplex oligo ligated into the digested product with Hi-T4 ligase (NEB #M2622) at RT for 1 hr. For each cloning reaction, assembled vectors were transformed into NEB Stable *E. coli* (NEB #C3040H) and grown at 32°C overnight. Single colonies were picked and miniprep (Qiagen #27104), and sequences were verified by Plasmidsaurus whole plasmid sequencing.

sgRNA library construction and quality control

Vectors pEM040 and pEM047 were sequentially digested with BstXI and BlnI, and the desired 8119 bp / 7973 bp products were gel purified from an 0.8% agarose gel. Each vector contains a 1133-bp LacZ fragment that is removed by double digestion with BstXI and BlnI, allowing visualization of complete digestion during gel purification. sgRNAs were chosen based on previous published and unpublished data demonstrating on-target activity, and NTC guides were selected that minimized differential expression in previous

screens. sgRNA libraries were ordered as IDT oPools with 5' overhang 5'- agtatcccttgaggagaaccacctgttg-3' and 3' overhang 5'-gtttaa gagtaagctggaacagcatagcaag- 3'. For library cloning, oPools were resuspended at 100 ng/ μ L, and 10 μ L (1 μ g) was used as input to a second strand synthesis reaction with 2x KAPA HiFi Master Mix (Roche, #07958935001) and an equimolar amount of primer P276R. (Table S6) The reaction was run in a thermal cycler with the following parameters: 95° 3', 64° 15" (ramp down 0.1°C/sec), 72° 10', 4° hold. After purification of the double-stranded product (NEB #T1030), the library was cloned into digested pEM040 and pEM047 via HiFi assembly (NEB #E2621) for 1 hr at 50°C, using a 5:1 molar ratio of insert library to vector. Electroporation was performed with an Eppendorf Eporator into Endura Electrocompetent Cells (LGC 60242-2) per manufacturer protocol. After electroporation, each reaction was incubated in 1 mL Recovery Media at 32°C for 1.5 hr, then expanded into 100 mL LB + carbenicillin for overnight culture. Plasmid libraries were produced via midiprep (Qiagen #12941).

For QC of the constructed libraries, sgRNA protospacers were amplified via 2x KAPA HiFi Master Mix (Roche, #07958935001) from midipreps with primers containing Illumina P5 and P7 adapters and Nextera i7 indices. (oJR232, oJR233, Table S6) Following 2x ProNex purification (Promega, NG2001), amplicons were sequenced on an Illumina MiSeq with PE70 read structure. Resulting fastq files were processed with a custom Python script to quantify representation of each sgRNA in the library (Figure S4).

Multome Perturb-seq: experimental procedure

For the Multome Perturb-seq experiment, populations of 1.2×10^7 RPE-1-ZIM3 cells were infected with lentivirus containing each library at a concentration of 400 μ L harvested lentivirus per million cells (targeting MOI 0.1) and plated into 3 T175 flasks each in the presence of 8 μ g/mL polybrene (Day 0). On Day 3, cells were sorted for GFP fluorescence by FACS (FacsSymphony S6, BD Biosciences). Infection titers of 6.6% and 9.4% were obtained for the +SIRLOIN/TSO vector and -SIRLOIN/TSO vector, respectively. 1.2×10^6 and 2.0×10^6 cells were recovered from sorting, and each population was plated in a 10cm dish. On Day 6, 1×10^6 cells from each population were prepared for encapsulation via nuclei isolation, following the 10x protocol "Nuclei Isolation for Single Cell Multome ATAC + Gene Expression Sequencing" (CG000365, Rev C) with 4 min lysis time (Step 2d) and verification of complete lysis by Trypan Blue staining and visualization on the Countess II FL (Life Technologies). We observed that digitonin quality is extremely important to high-quality nuclei extraction in RPE-1 cells; digitonin that is expired or not thawed with rapid mixing at 65°C before use will result in incomplete lysis and affect the quality of extracted nuclei.

Isolated nuclei were resuspended in 500 μ L PBS + 1 U/ μ L Sigma Protector RNase inhibitor (Sigma #3335402001), and 25 μ L 7-AAD (Thermo #00-6993-50) was added to stain intact nuclei. Samples were immediately sorted by FACS (FacsSymphony S6, BD) to purify 7-AAD+ nuclei and remove debris, resulting in yields of 409,591 nuclei (+SIRLOIN/TSO) and 404,204 nuclei (-SIRLOIN/TSO). While sorting nuclei is an optional step of the 10x Multome protocol, we observed that sorting for intact nuclei reduced ambient RNA which could impede ID assignment. Sorted nuclei were resuspended in 10x Nuclei Buffer targeting a concentration of 4000 nuclei/ μ L, assuming 30% loss of the sorted yield during resuspension. Targeting 10,000 nuclei per reaction and accounting for the 10x-indicated recovery factor of 1.61, we loaded 4 μ L ($(10,000 \text{ nuclei} \times 1.61) / 4000 \text{ nuclei}/\mu\text{L}$) of each nuclei stock into transposition reactions per 10x protocols (Chromium Next GEM Single Cell Multome ATAC + Gene Expression User Guide, Rev F), and proceeded with GEM generation, barcoding, and cleanup using the Chromium Controller and Chromium Next Gen Chip J (10x Genomics #1000283) with no protocol modifications. ATAC libraries were constructed with no protocol modifications and 7 PCR cycles for ATAC library amplification (Step 5.1d). cDNA amplification was performed with no protocol modifications and 6 PCR cycles (Step 6.1d). Gene expression libraries were constructed with no protocol modifications, using 12 PCR cycles for library amplification (Step 7.5d). After PicoGreen quantification and quality control by Agilent TapeStation, libraries were pooled to equimolar concentrations and sequenced to a target depth of 20,000 reads per cell on an Illumina NovaSeq X in a PE28/88 run using the NovaSeq X 10B or 25B Reagent Kit (100 or 300 Cycles) (Gene Expression) or to a target depth of 25,000 reads per cell on a NovaSeq 6000 in a PE50 run using the NovaSeq 6000 S1, S2, or S4 Reagent Kit (100 or 200 Cycles) (ATAC). The loading concentration was 1nM (NovaSeq X) or 0.6-1.2nM (NovaSeq 6000) and a 1% spike-in of PhiX was added to each run for quality control purposes.

Multome Perturb-seq: T7-assisted library prep

A full step-by-step protocol is available in [Methods S1](#).

To construct perturbation identity transcript libraries from pooled cDNA, 10 μ L (25% of final yield by volume) of amplified cDNA from each library (Chromium Next GEM Single Cell Multome ATAC + Gene Expression User Guide Rev F, Step 6.2n product) was *in-vitro* transcribed (IVT) with the NEB Quick T7 HiScribe Kit (NEB #E2050) in 25 μ L reactions per manufacturer protocol. IVT reactions were incubated at 37°C for 2 hr. 3.8 μ L dH₂O and 1.2 μ L DNase I were then added directly to each reaction (30 μ L total reaction volume), and the reactions were incubated at 37°C for 15 min. IVT RNA product was purified with 1.8x RNAClean (Beckman Coulter # A63987) and eluted in 16.5 μ L RNase-free dH₂O. 1 μ L product was used to quantify RNA yield with the Qubit RNA BR kit. The entire purified IVT product was used as template for reverse transcription with the Maxima H Minus First Strand cDNA Synthesis Kit (Thermo # K1652) as follows. On ice, 2 μ L 10 μ M RT primer (T7_TruseqRead1, Table S6) and 1.25 μ L 10 mM dNTPs were added to 15.5 μ L IVT product. The reactions were incubated at 65°C for 5 min and immediately returned to ice. To each reaction on ice, 5 μ L 5x RT buffer and 1.25 μ L RT Enzyme Mix were added. Reactions were incubated at 50°C for 30 min and heat-inactivated at 85°C for 5 min. Next, the entire RT product was used directly in index PCR across 5 50 μ L reactions to ensure that the volume of unpurified RT product did not exceed 10% of each individual PCR reaction. A 5x PCR master mix for each sample was built on ice as follows: to 25 μ L unpurified RT product, we added 97 μ L dH₂O, 1.5 μ L 100 μ M P065-P5 (P5 reverse primer), 1.5 μ L 100 μ M P065-N7xx (indexed P7 forward primer, captures U6 promoter directly 5' of protospacer), and 125 μ L 2x KAPA HiFi Master Mix (Roche #07958935001).

(Table S6) The master mix was divided among 5 50- μ L reactions in a PCR strip, and PCR was performed with the following parameters: 95° 3', 8 cycles of [98° 15", 70° 20"], 72° 1', 4° hold. PCR reactions were pooled and purified via 1.15x ProNex (+SIRLOIN/TSO) or 1.2x ProNex (-SIRLOIN/TSO). Yield was measured with the Qubit dsDNA HS kit (Thermo #Q32851) and amplicons were visualized as single peaks at 584 and 516 bp respectively on a Bioanalyzer (Agilent) using the High Sensitivity DNA Chip. After PicoGreen quantification and quality control by Agilent TapeStation, libraries were pooled to equimolar concentrations and sequenced to a target depth of 3,000 reads per cell on an Illumina NovaSeq X in a PE28/88 run using the NovaSeq X 10B or 25B Reagent Kit (100 or 300 Cycles). The loading concentration was 1nM and a 1% spike-in of PhiX was added to the run for quality control purposes.

Multiome Perturb-seq: PCR enrichment CROP-seq library prep

ID transcripts were isolated from Gene Expression library (Chromium Next GEM Single Cell Multiome ATAC + Gene Expression User Guide Rev F, Step 7.6q product) for comparison with T7-assisted library prep by traditional CROP-seq. Transcripts were amplified from 200 ng library product across 5 50- μ L PCR reactions with 2x KAPA HiFi Master Mix (Roche, #07958935001), P065-P5, and P065-N7xx (Table S6). PCR was performed with the following parameters: 95° 3', 20 cycles of [98° 15", 70° 10"], 72° 1', 4° hold. Reactions were pooled and purified with 1.1x (+SIRLOIN/TSO) or 1.15x (-SIRLOIN/TSO) ProNex. We reduced ProNex ratios slightly vs. T7 library prep to reduce the presence of starting material in final libraries, a known issue with CROP-seq. Yield was measured with the Qubit dsDNA HS kit (Thermo #Q32851) and amplicons were visualized as single peaks at 584 and 516 bp respectively on a Bioanalyzer (Agilent) using the High Sensitivity DNA Chip. After PicoGreen quantification and quality control by Agilent TapeStation, libraries were pooled to equimolar concentrations and sequenced on a NovaSeq X in a PE28/88 run using the NovaSeq X 10B or 25B Reagent Kit (100 or 300 Cycles). The loading concentration was 1nM and a 1% spike-in of PhiX was added to the run for quality control purposes. While we targeted the same sequencing depth as the T7 libraries, the presence of gene expression library starting material in the final CROP-seq library resulted in fewer ID reads at the same total sequencing depth. Of 30 million target reads, we recovered 8.33 million (+SIRLOIN/TSO) and 13.8 million (-SIRLOIN/TSO).

RT-qPCR characterization of CROP-seq vectors

Leftover sorted nuclei and whole cells from the Multiome Perturb-seq experiment were used to quantify ID transcript expression. Total RNA was extracted from each sample (+/- SIRLOIN/TSO, nuclei and whole cells, 4 samples total) with the NEB Total RNA Miniprep kit (NEB #T2010). 10 ng nuclear RNA and 100 ng whole-cell RNA was used as template for RT-qPCR using the NEB Luna Universal One-Step RT-qPCR kit (NEB #E3005). PCR was performed per manufacturer recommendations, though the RT temperature was raised to 60°C. We targeted two 250 bp amplicons in the ID transcript for consistency, one within the coding sequence of GFP and the other within the U6 promoter. The MALAT1 nuclear RNA was used as a normalization control, similarly to the original characterization of SIRLOIN elements.²² Ct values were measured on a CFX96 Real-Time System (Bio-Rad), and Δ Ct was calculated as the difference between amplicon and MALAT1 Ct values in each condition. $\Delta\Delta$ Ct was calculated by subtracting the Ct values of the +SIRLOIN/TSO samples from that of the -SIRLOIN/TSO samples, and expression fold change for each amplicon in the +SIRLOIN/TSO condition was calculated as $2^{-\Delta\Delta Ct}$. Standard errors were propagated through Ct calculations as previously described.⁵⁹

Flow cytometry validation of on-target knockdown

1.5×10^5 RPE-1-ZIM3 cells were transduced with lentivirus containing CD81 or NTC sgRNAs in +SIRLOIN/TSO and -SIRLOIN/TSO CROP-seq vectors to a target MOI of 0.1 via plating with 8 μ g/ml polybrene (Day 0). Transduced cells were sorted for GFP positivity on Day 3. On Day 6, cells were harvested, washed with PBS, and resuspended in 180 μ L staining buffer (BD #554657). 20 μ L CD81-APCH7 antibody (BD #656154) was added, and cells were incubated protected from light at 4°C for 15 min. Cells were then washed with 1 ml PBS and resuspended in 200 μ L staining buffer, and CD81 signal was analyzed on an Attune CytPix flow cytometer (Thermo).

Direct-capture Perturb-seq experiment in single nuclei

For the direct-capture Perturb-seq experiment to benchmark the assignment rate of Multiome Perturb-seq, RPE-1-ZIM3 cells were transduced at low MOI with a direct-capture lentiviral vector (pRCA360) encoding the RPE-1 Essentials CRISPRi library described in our concurrent work.⁶⁰ All experimental parameters were consistent with the Multiome screen, including nuclei preparation and sorting. As the 10x Nuclei Buffer used in Multiome experiments is not supplied with 3' GEX kits, nuclei were resuspended in PBS with 0.04% BSA and 1 U/ μ L RNase inhibitor after sorting. Encapsulation, cDNA generation, and Gene Expression and Feature Barcode library prep were performed according to the 10x Chromium Next GEM Single Cell 3' Reagent Kits v3.1 Dual Index with Feature Barcode Protocol (CG000389, Rev C) with no modifications. sgRNA assignments were called after cellranger count and cellranger aggr with a final threshold of 5 UMI to account for failed thresholding by the cellranger pipeline.

QUANTIFICATION AND STATISTICAL ANALYSIS

sgRNA identity calling

For analysis, sgRNA assignments were called on the T7 +SIRLOIN/TSO library with deeper sequencing using a custom Python script, `cropseq_call_guides_multiome.py`. Briefly, cellranger-7.1.0 was run on ID library reads, associating sgRNA-containing reads with 10x cell barcodes and UMIs (cellranger count -chemistry=ARC-v1). Based on cells called by cellranger-arc (below), reads were

filtered to those associated with cell-containing droplets and aligned to the protospacer library with fuzzy string matching (rapidfuzz), allowing at most a Levenshtein distance of 1 away from expected sgRNAs for a protospacer read to be considered for assignment. Reads were collapsed by cell barcode and UMI, creating a dataframe of the ID transcripts in each cell (one of 16 options based on alignment to the protospacer library) labeled with UMI and total read counts. The relationship between UMI and total read counts was used to separate true ID transcripts from PCR artifacts as follows. First, UMI and total read counts were $\log_2$ -transformed, and possible “Top IDs” representing true assignments with $\log_2(\text{UMI}) \geq 2.75$ were selected. An HDBSCAN model (min_samples = 10, metric = ‘l1’, cluster_selection_epsilon = 0.5) was fit to Top IDs to further filter these putative true assignments. Minority clusters and IDs categorized as noise (label -1) were removed, and the cluster with higher average UMI was selected as representative of true assignments. A logistic regression model was trained on all Top IDs, then applied to the entire dataset of ID transcripts to infer membership in the true assignment cluster. IDs for which the model predicted membership in this cluster were kept as true assignments, and other IDs were removed. This procedure results in a clear plane of separation between called and non-called transcripts (Figure S2), and removes likely PCR chimeras (e.g., transcripts for which the UMI-total read relationship deviates significantly from confident assignments, such as transcripts with a single UMI and read representing a likely template-switching event during index PCR). We additionally filtered the model predictions such that no ID transcript with less than 3 UMI would be called, regardless of its cluster prediction.

For comparisons of assignment between library prep and vector conditions, we downsampled individual reads from the above dataframe of ID transcripts (pd.DataFrame.sample()) before collapsing to UMI and total read counts, such that the total number of reads in this dataframe divided by the number of cells called by cellranger was constant (850 reads per cell, based on the minimum common sequencing depth available across conditions). We ran the ID assignment script as above, but changed the “Top ID” $\log_2(\text{UMI})$ threshold to 2 (T7) or 1.75 (CROPseq).

Single-cell data processing and QC filtering

cellranger-arc-2.0.2 (10x Genomics) was used for alignment of Gene Expression and ATAC libraries to the transcriptome and genome respectively (reference: cellranger-arc-GRCh38-2020-A-2.0.0), using cellranger-arc count with default parameters. For gene expression processing, sgRNA ID assignments were merged with cellranger gene expression outputs in Scanpy,⁵⁷ and quality control metrics were calculated with sc.pp.calculate_qc_metrics(). We inspected these metrics by sgRNA assignment status (Figure S4) and filtered low-quality cells based on total UMI and library complexity (number of unique genes). Mitochondrial genes were removed, and cells were filtered to those with single sgRNA assignments. The resulting AnnData object was further filtered using the perturbseq codebase³ to remove genes expressed in fewer than 5% of cells with strip_low_expression(threshold = 0.05), and counts were then z-normalized to control by normalizing total counts in all cells to median UMI, subtracting the NTC mean of each gene, and dividing by the NTC standard deviation. The resulting normalized matrix was read back into Scanpy to regress out cell cycle genes (obtained from Harmonizome⁶¹).

For ATAC processing, 10x fragment files were converted to tile matrices with SnapATAC2.⁵⁸ QC metrics were calculated, and cells with ≤ 1000 unique fragments were removed. From the genome-wide tile matrix, 2.5×10^5 features were selected for downstream analyses based on accessibility in all cells. The combined quality control process applied to both modalities left 4,724 high-quality nuclei with single sgRNA assignments and complete expression and accessibility profiles (Table S2).

Identification of differentially expressed genes and differentially accessible peaks

Differentially expressed genes were called on the filtered z-normalized matrix described above with a Kolmogorov-Smirnov test (perturbseq.ks_de()). *P*-values were corrected for false discovery rate via the Benjamini-Hochberg procedure, and genes with FDR-corrected $q < 0.1$ were considered differentially expressed. Differentially accessible ATAC peaks were identified using SnapATAC2’s suggested workflow. From the filtered tile matrix described above, peaks were called with MACS3⁶² (snap.tl.macs3()) on pseudobulk accessibility profiles quantified with paired-insertion counting^{63,64} for each perturbation target and NTC, and a cell-by-peak matrix with fixed-width, non-overlapping features was created with snap.tl.merge_peaks() followed by snap.tl.make_peak_matrix(counting_strategy = ‘paired-insertion’). For each perturbation, differentially accessible peaks were identified with a Mann-Whitney test (scipy.stats) between counts in NTC and perturbed cells, performed on each feature of the cell-by-peak matrix for which any perturbed or NTC cells had at least one count. *P*-values were corrected for false discovery rate via the Benjamini-Hochberg procedure, and peaks with FDR-corrected $q < 0.1$ were considered differentially accessible. Log fold change values were calculated as the $\log_2$ -transformed ratio of mean counts in perturbed cells to control cells. A pseudocount of 1×10^{-3} was added to the numerator and denominator during this calculation to avoid infinite values. To identify concordant differential changes, each peak was labeled with its nearest gene, and matches were considered concordant if both the peak and its nearest gene were differentially expressed or accessible with $q < 0.1$ and the sign of the gene expression z-score was equal to the sign of the peak log fold change.

Annotation of ATAC-seq peaks with ENCODE cis-regulatory elements

A bed file of cis-regulatory elements was downloaded from ENCODE SCREEN.⁶⁵ To label ATAC differential features, PyRanges was used to intersect this bed file with the coordinates of fixed-width merged peaks composing the features of the cell-by-peak matrix described above. Peaks were labeled with CREs based on overlap of at least 1bp. To label total peaks for Table S4, unique peaks from MACS3 output were intersected and labeled with the regulatory element for which the peak had the largest overlap.

For a finer-grained characterization of promoters, we supplemented promoter annotations with a database of human promoters downloaded from the Eukaryotic Promoter Database,⁶⁶ and we extended coordinates by 950 bp upstream to represent promoter regions. Features annotated as “prom” overlap a promoter region. The SOX5 enhancer peak shown in [Figures 2I](#) and [2J](#) was labeled as overlapping a promoter region, but contained a proximal enhancer in the ENCODE dataset. We characterize the peak as an enhancer based on its ENCODE annotation to maintain consistency, but the feature remains labeled as overlapping a promoter region in the raw data (e.g., in [Table S5](#)).

Clustering and visualization of expression and accessibility profiles

Gene expression profiles were clustered by decomposing the processed single-cell expression matrix with PCA into 10 components. Component values were grouped by perturbation identity and averaged, and these pseudobulk embeddings were hierarchically clustered using 1 - Pearson correlation as the distance metric and Ward’s linkage. ATAC profiles were clustered by decomposing the processed single-cell tile matrix with SnapATAC2’s spectral embedding algorithm into 11 components. We removed the first component of this embedding, finding that it correlated strongly with the number of unique fragments recovered in each single cell. The remaining 10 components were similarly grouped by perturbation identity and averaged, and the pseudobulk spectral embeddings were clustered by hierarchical clustering with 1 - cosine similarity as the distance metric and Ward’s linkage. For clustermap generation in each case, we used `hc.linkage(method = ‘ward’, optimal_ordering = True)` from `scipy.cluster.hierarchy`. For UMAP visualizations ([Figure S5](#)), the above 10-component PCA embedding (expression) or spectral embedding (ATAC) was reduced to 2 dimensions with `UMAP(n_neighbors = 8, min_dist = 0.25, random_state = 8)` using correlation as the distance metric for expression and cosine distance for accessibility. The joint UMAP visualization was constructed by concatenating the 10-component PCA and spectral embeddings for expression and accessibility and applying the UMAP transformation as above using cosine distance.

Sparse PCA co-embedding of expression and accessibility features

To organize perturbation effects and decompose peak-driven vs. gene-driven changes, we adapted a sparse PCA approach used by our group to interpret Perturb-seq data to the Multiome setting. For this analysis, we first z-normalized paired insertion counts within the peak matrix as follows: from the cell-peak matrix described above, features were pre-filtered to remove the lower 25% based on count values in all cells and any peaks more than 50 kb away from the nearest gene. Means and standard deviations of counts in NTC cells for each feature were then generated, and the filtered peak matrix was z-normalized by subtracting the NTC mean and dividing by the NTC standard deviation. From the z-normalized peak matrix, we then removed peaks that were not differentially accessible in any perturbations, resulting in 5,060 peaks. We similarly z-normalized gene expression matrices as described above, and removed genes not differentially expressed in any perturbations. This resulted in a concatenated expression/accessibility matrix of 5,060 peaks and 7,885 genes for downstream analysis.

Our sparse PCA model was constructed by modifying sklearn’s SparsePCA object to constrain feature weights to be nonnegative, using `dict_learning(positive_code = True)`. This procedure promotes interpretability by 1) limiting the number of gene and peak features within each resulting program, as the model is fit with a regularization term penalizing nonzero feature weights; and 2) ensuring that features within a program are co-regulated in the same direction by each perturbation, via the nonnegativity constraint. We fit a sparse PCA model to the above concatenated matrix, deciding empirically on the number of programs to fit (limited by the low-rank structure of the matrix, with only 13 perturbations and 3 underlying complexes) and the value of the regularization parameter alpha. Fitting 8 programs with alpha = 0.25 resulted in expected clustering of perturbation weight vectors into known complexes, with a sparsity level that allowed for substantial gene-peak heterogeneity between programs while not sacrificing the value of the method as a feature selection tool. Peak features in each program were then labeled with their nearest gene and ENCODE cis-regulatory element annotations based on overlap, if applicable. Features, weights, and annotations for each program are available in [Table S5](#). SOX5 target genes were obtained from Harmonizome.⁶¹

Mapping of sparse programs to in vivo retinal atlas data

Preprocessed snRNA-seq data from the Zuo et al.³⁴ retinal atlas was downloaded from CellxGene (<https://cellxgene.cziscience.com/collections/5900dda8-2dc3-4770-b604-084eac1c2c82>). The UMAP representation of the dataset colored by cell type was reproduced from data in the preprocessed AnnData object. To calculate the Program 2 score for each single cell, we intersected the top 50 features in Program 2 with the preprocessed expression matrix, removing unannotated features. For the remaining 43 features, we calculated the dot product between the Program 2 feature weights and the preprocessed log-normalized expression counts to produce a scalar score for each cell. These scores were z-normalized and plotted on the pre-existing UMAP representation of the dataset. Marker genes of each neuronal cell type ([Figures 2E](#), [2M](#), and [S8](#)) were obtained from [Data S3](#) of Zuo et al. by filtering to significantly differential genes (FDR-controlled $q < 0.1$) and gene score > 0 .

Generation and visualization of ATAC-seq coverage tracks

Bigwig coverage tracks were generated at pseudobulk level for each perturbation and NTC cells as follows. The bam output of cellranger-arc was filtered with pysam to include only reads with the ‘CB’ tag corresponding to perturbation-assigned cell barcodes, creating a pseudobulk.bam file for each perturbed subpopulation and NTC. These files were indexed with samtools index

and converted to bigwig coverage tracks with deepTools bamCoverage –normalizeUsing RPGC –effectiveGenomeSize 2913022398 –binSize 20. Tracks were visualized with pyGenomeTracks.⁶⁷

Construction of pseudotime trajectories

We applied a diffusion pseudotime approach as previously described⁴⁴ to create an ordered embedding of control and perturbed single cells. We constructed a new gene expression matrix in scanpy by normalizing each cell to equal total UMI and log1p-transforming normalized counts, avoiding z-normalization to maintain meaningful values for NTC cells. Similarly, starting with the cell-by-peak matrix as above, we filtered peaks more than 50 kb away from the nearest gene and tf-idf normalized raw paired insertion count values. We then constructed an individual pseudotime trajectory for each perturbation as follows. We selected only features that were differentially expressed or accessible in the perturbation and concatenated these features into a combined expression and accessibility matrix. For many perturbations, the target gene and promoter were differentially expressed and accessible (Figure 1); we removed these genes and peaks from the calculation to maintain on-target expression and accessibility as a “positive control” to evaluate the fidelity of our embedding in separating perturbed and control cells. We then scaled each feature to zero mean and unit variance and decomposed the matrix into 10 principal components. We constructed the nearest-neighbors graph on this PCA embedding using sc.pp.neighbors(metric = ‘correlation’), and constructed a diffusion map with sc.tl.diffmap(). To set a root cell for diffusion pseudotime construction (required by scanpy’s implementation), we identified the diffusion component with the highest absolute-value Pearson correlation with a binary perturbed/control variable for each single cell, and chose the cell at the appropriate extreme of this component (lowest value if correlation was negative, highest value if positive) as the root cell.

We then constructed the pseudotime with sc.tl.dpt(), and visualized cells ordered by their pseudotime rank via a UMAP embedding fit on the above concatenated differential expression / accessibility matrix. Pseudotime ranks aligned well with perturbed/control identities for most perturbations. Notably, alternative trajectories that do not require the *a priori* choice of a root cell (single-component spectral embedding, linear discriminant analysis) produced indistinguishable results when applied to the combined population. Visualizations (Figure 2J) were constructed by smoothing ordered single-cell expression and accessibility values with a 500-cell rolling average, then scaling each feature separately to a range of 0–1.

AutoGluon modeling of expression levels for program and control genes

We applied the AutoGluon Tabular AutoML framework to model expression as a function of *cis*-chromatin context, fitting separate regression models for each program gene and control gene. We selected 457 annotated protein-coding genes with top-100 feature weights in each sparse program as “program genes.” Control genes were selected from genes not differentially expressed in any perturbation condition and not appearing as a peak feature in any program. As program genes are disproportionately highly expressed, random sampling of control genes resulted in a distribution of absolute expression levels that was not matched to program genes, potentially confounding the comparison of modeling performance. To correct this, we augmented an initial list of randomly sampled control genes via resampling weighted on mean expression level. We fit a Gaussian kernel density estimate (scipy.stats.gaussian_kde()) to the upper tail of the program gene expression distribution, then evaluated the KDE on the mean expression levels of control genes with $\log(1 + \text{TPM}) > 2$. The square root of the empirical probability density for each control gene was then used as a sampling weight. By inspection, replacing the lower 30% of randomly sampled control genes with higher-expression control genes resampled in this way produced similar distributions (Figure S7A).

To mitigate ATAC sparsity at the single-cell level, we clustered single cells in each perturbation condition and control into “metacells” via k-means ($k = 10$) performed on the spectral embedding of the preprocessed ATAC tile matrix (described above), such that the model predicts expression for each metacell rather than for each single cell. This clustering procedure resulted in a total of 140 metacells (13 perturbations and control with 10 clusters each). For modeling, the design matrix was constructed from the ATAC tile matrix by subsetting to the tiles nearest the start and end coordinates of each gene, then adding a 250 kb (500 tile) window on both 5’ and 3’ ends. Counts in each tile normalized to reads per million for each single cell were then averaged across each metacell. We found that further feature scaling (i.e., with z-score or min/max) did not improve modeling performance, consistent with the scale invariance of the tree-based models employed by AutoGluon. To construct the response variable, the raw gene expression matrix was normalized to equal UMI per single cell, and counts (transcripts per 10k) were averaged across each metacell. An 80%/20% train-test split was created with scikit-learn’s train_test_split(random_state = 42), stratified on perturbation identity. We modeled the expression level of each gene as a percentage of the maximum observed in the training set, using scikit-learn’s MinMaxScaler() to fit and transform the training set expression to a 0–1 scale, then transforming the test set expression to this scale at evaluation time. We then trained AutoGluon TabularPredictor() objects for each gene using MSE as the objective. To train 914 gene-level models, we arrived at a tradeoff between predictive performance and wall time / resource intensity by disabling AutoGluon’s bagging and dynamic stacking (presets = ‘medium_quality’) but employing its optimized zero-shot portfolio of hyperparameters (hyperparameters = ‘zeroshot’). After observing low performance, PyTorch neural networks were removed to further accelerate training. Models were evaluated by calculating the Spearman correlation between observed and predicted expression levels on test metacells.

Feature importance for each chromatin tile was evaluated on the test set by permutation-shuffling with AutoGluon’s feature_importance() method, producing a score for each tile quantifying the decrease in predictive performance observed by leaving out

each feature along with an associated p-value. Feature p-values were corrected for false discovery by the Benjamini-Hochberg procedure. For analysis of regulatory elements overlapping important features, a bed file of ENCODE cCREs (described above) was intersected with features for each gene using PyRanges. Features that overlapped more than one cCRE were labeled with the cCRE with higher overlap. Top features were defined as tiles with importance scores at or above the 90th percentile across all genes with FDR-corrected $q < 0.1$. For visualization of accessibility and feature importance in [Figure S8](#), ATAC-seq coverage tracks were constructed as described above, and pyBigWig was used to plot values.